Module/Pathway/Taxon Review: Bacterial Lipoprotein Maturation in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Pathway query:** UPA00665 (UniPathway "Protein modification; lipoprotein biosynthesis") **Module:** Ordered maturation of lipobox prolipoproteins — Lgt → LspA → Lnt **Date:** 2026-07-14

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1. Executive summary

The canonical three-step diderm lipoprotein-maturation route is **fully present and satisfiable** in *P. putida* KT2440. All three steps map to single-copy, reviewed Swiss-Prot genes:

Step	Enzyme	Gene	Locus	UniProt	EC	KO
1. Diacylglyceryl transfer	Lgt	*lgt* (*umpA*)	PP_5142	Q88CN8	2.5.1.145	K13292
2. Signal-peptide cleavage	LspA	*lspA*	PP_0604	Q88Q91	3.4.23.36	K03101
3. N-acylation	Lnt	*lnt*	PP_4790	Q88DN4	2.3.1.269	K03820

The local candidate list contained **only** *lspA* (PP_0604). This is a **KEGG-boundary artifact**, not a genome gap: only LspA's KO is placed in KEGG's Protein export map (ppu03060), whereas *lgt* and *lnt* are unmapped in that KEGG pathway even though both genes exist. **Recommendation:** **`module_needs_revision`** — add PP_5142 (Lgt) and PP_4790 (Lnt) as covered steps. No step should be marked `gap` or `not_expected_in_target_taxon`.

Confidence is **high by orthology** (specific InterPro family calls, UniPathway sub-step annotations, single-copy loci) but there is **no direct experimental characterization in KT2440** — all three entries are protein-existence level 3 (inferred from homology). The strongest nearby experimental anchor is the *P. aeruginosa* LspA crystal structure (P 26912896).

To upgrade this homology-only evidence, I verified at the sequence level that **each KT2440 enzyme conserves every experimentally-defined catalytic residue** of its characterized homolog (Section 4.4), which raises functional confidence to high and specifically clears the Lnt over-annotation flag.

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2. Target-organism pathway definition

Included process. Post-translational maturation of lipobox-containing prolipoproteins at the cytoplasmic (inner) membrane, in strict order: 1. **Lgt** transfers a diacylglyceryl group from phosphatidylglycerol onto the thiol of the invariant lipobox cysteine, forming diacylglyceryl-prolipoprotein. 2. **LspA** (signal peptidase II), an aspartyl peptidase, cleaves the signal peptide immediately N-terminal to the modified cysteine, yielding apolipoprotein (N-terminal S-diacylglyceryl-cysteine). 3. **Lnt** N-acylates the free α-amino group of that cysteine, producing the mature **triacylated** lipoprotein.

UniPathway mapping. UniProt annotates the three KT2440 enzymes with the identical pathway "Protein modification; lipoprotein biosynthesis" and the three ordered sub-steps *(diacylglyceryl transfer)*, *(signal peptide cleavage)*, *(N-acyl transfer)* — i.e., the query **UPA00665** maps exactly to this module and is satisfied.

Neighboring pathways to keep separate. - **Lol sorting/trafficking** (LolABCDE): outer-membrane targeting of mature lipoproteins. KT2440 encodes LolA (PP_4003), LolB (PP_0724), LolC (PP_2154), LolD (PP_2155), LolE (PP_2156). These are **downstream** and belong to a distinct module. - **General Sec/Tat protein export** (KEGG map03060): the broad overview map that contains LspA but is not lipoprotein-specific; do not collapse the maturation module into it. - Substrate-specific lipoprotein functions (e.g., Lpp/OprI-type, LptE, BamB/C/D/E) are downstream clients, not maturation steps.

Alternate names / DB definitions. Lgt = *umpA*; LspA = "prolipoprotein signal peptidase / SPase II / signal peptidase II"; Lnt = "apolipoprotein N-acyltransferase / ALP N-acyltransferase." EC set: 2.5.1.145 / 3.4.23.36 / 2.3.1.269.

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3. Expected step model

For a **diderm (Gram-negative) gammaproteobacterium** such as KT2440, all three steps are **expected and essentially non-optional**:

- Lgt is the gating step and *lgt* deletion is lethal in most Gram-negatives (P 26729647).
- The route requires sequential Lgt → LspA → Lnt activity (P 28698362).
- Lnt-dependent N-acylation (triacylation) is characteristic of proteobacteria. Unlike many monoderm Firmicutes (which frequently lack a canonical Lnt and produce diacylated lipoproteins or use lineage-specific N-acyltransferases), KT2440 is expected to carry the canonical Lnt — consistent with the present PP_4790.

Ordering constraints in the generic module (Lgt precedes LspA precedes Lnt) hold in KT2440.

Substrate pool. The pathway is functionally engaged, not vestigial: UniProtKB annotates **≥76 lipoprotein (KW-0449) proteins** in the KT2440 proteome (52 also carrying a signal peptide), i.e. an abundant client set of lipobox prolipoproteins requiring maturation — consistent with the essentiality of the machinery.

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4. Candidate genes and evidence

4.1 LspA — PP_0604 / Q88Q91 (the sole metadata candidate)

- **Role:** signal peptidase II; cleaves signal peptide from diacylglyceryl-prolipoprotein.
- **Evidence type:** homology-inferred (PE3, HAMAP MF_00161). InterPro **IPR001872 "Peptidase A8, signal peptidase II"** (specific); Pfam PF01252. GO:0004190 (aspartic-type endopeptidase), GO:0006508, GO:0005886.
- **Species transfer: strong.** The *P. aeruginosa* LspA–globomycin crystal structure identifies LspA as an aspartyl peptidase (P 26912896); *P. aeruginosa* is a close congener, so mechanism/active-site transfer to KT2440 is well

justified. LspA is a validated antibacterial target (globomycin/myxovirescin; PMIDs 32768648, 22232277).

- **Caveats:** GO:0004190 "aspartic-type endopeptidase activity" is a broad parent term; the specific IPR001872 family and EC 3.4.23.36 anchor the correct function.

4.2 Lgt — PP_5142 / Q88CN8 (missing from metadata, present in genome)

- **Role:** phosphatidylglycerol–prolipoprotein diacylglyceryl transferase; step 1.
- **Evidence type:** homology-inferred (PE3, HAMAP MF_01147). InterPro **IPR001640 "... diacylglyceryl transferase Lgt"** (specific); Pfam PF01790. GO:0008961 (specific activity), GO:0042158, GO:0005886. UniPathway sub-step "(diacylglyceryl transfer)."
- **Species transfer:** strong by orthology; direct structural/mechanistic basis from *E. coli* Lgt (P 26729647) and cross-phylum conservation of Lgt (P 7592473).
- **Caveats:** none major; single-copy, specific family. Gene lies at ~5.87 Mb (complement 5867757–5868563), separate locus from the others.

4.3 Lnt — PP_4790 / Q88DN4 (missing from metadata, present in genome)

- **Role:** apolipoprotein N-acyltransferase; terminal step, produces triacylated lipoprotein.
- **Evidence type:** homology-inferred (PE3, HAMAP MF_01148). InterPro **IPR004563 "Apolipoprotein N-acyltransferase"** (specific family) plus generic **IPR003010 "Carbon-nitrogen hydrolase"** domain; Pfam PF00795 + PF20154. GO:0016410 (N-acyltransferase), GO:0042158, GO:0005886. UniPathway "(N-acyl transfer)."
- **Species transfer:** strong by orthology; mechanistic framework from structural work on Lnt (P 28698362).
- **Caveats:** PF00795/IPR003010 is the **nitrilase / carbon-nitrogen-hydrolase superfamily**, which harbors many unrelated activities — a generic over-propagation flag in isolation. Here it is **mitigated** by the dedicated IPR004563 family assignment and EC 2.3.1.269, so the Lnt call is reliable.

Paralog check: proteome scans on each EC (2.5.1.145, 3.4.23.36, 2.3.1.269) returned exactly one hit — **no paralog ambiguity** for any step.

4.4 Catalytic-residue conservation (sequence-level verification)

To move beyond database homology, KT2440 enzymes were pairwise-aligned to their characterized homologs and active-site residues checked:

Enzyme	Reference (characterized)	% identity	Catalytic residues checked	Result
LspA (Q88Q91)	*P. aeruginosa* PAO1 LspA (Q9HVM5; crystallized, P 26912896)	**78.7%**	Asp124, Asp143 (aspartyl dyad)	**both conserved** (near-identical context GHVVD**F**ILV / FN[V/L]AD**S**AIT)
Lgt (Q88CN8)	*E. coli* K12 Lgt (P60955; P 26729647)	55.1%	Arg143, Arg239 (essential)	**both conserved**
Lnt (Q88DN4)	*E. coli* K12 Lnt (P23930; P 28698362)	43.1%	Glu267, Lys335, Cys387 (nitrilase-superfamily triad)	**all conserved**

Conservation of the full catalytic set — including the Lnt catalytic Cys387 — raises functional confidence to high despite the PE3 (homology-inferred) status, and decisively refutes the possibility that PP_4790 is a mis-annotated CN-hydrolase-superfamily paralog. The 78.7% identity of LspA to the crystallized *P. aeruginosa* enzyme makes mechanistic species transfer strong.

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5. Gaps, ambiguities, and likely over-annotations

- **No true gaps.** Every expected step has a specific-family, single-copy gene.
- **Metadata gap is artifactual.** Lgt and Lnt are absent from the candidate list only because KEGG's ppu03060 pathway map does not contain their KOs (K13292, K03820); this is a database-boundary artifact.
- **Over-annotation flags (low risk):**
- LspA GO:0004190 is a broad endopeptidase parent term — acceptable given IPR001872/EC anchoring.

- Lnt CN-hydrolase superfamily membership (PF00795/IPR003010) is the classic over-propagation trap; here countered by IPR004563 **and** by conservation of the catalytic triad Glu267/Lys335/**Cys387** (Section 4.4) — flag cleared.
- **Evidence caveat:** all three are PE3 (inferred from homology) with no KT2440-specific wet-lab data; direct evidence exists only in relatives (*P. aeruginosa* LspA; *E. coli* Lgt/Lnt). However, sequence-level conservation of all experimentally-defined catalytic residues (Section 4.4) substantially mitigates this caveat.

6. Module and GO-curation recommendations

Module step	Call	Gene(s)
1. Diacylglyceryl transfer (Lgt)	**covered**	PP_5142 / Q88CN8
2. Signal-peptide cleavage (LspA)	**covered**	PP_0604 / Q88Q91
3. N-acylation (Lnt)	**covered**	PP_4790 / Q88DN4

- **Overall module:** `module_needs_revision` — expand the KT2440 candidate set from 1 gene (lspA only) to all three; the genome fully satisfies UPA00665.
- **Boundaries:** correct as drawn. Keep Lol sorting (LolABCDE: PP_4003/PP_0724/PP_2154/PP_2155/PP_2156) and general Sec/Tat export (map03060) as separate modules.
- **GO/annotation:** no new GO term requests needed. Optionally prefer the specific child terms already present (GO:0008961 for Lgt; retain EC/InterPro anchors for LspA and Lnt) over broad parents. Note lolC/lolE are TrEMBL (unreviewed) if the Lol module is later curated.
- **No `not_expected_in_target_taxon` calls** — the diderm gammaproteobacterial context expects all three steps, including canonical Lnt.

7. Genes to promote to full `fetch-gene` review

1. **PP_5142 (lgt / Q88CN8)** — highest priority; must be added to satisfy step 1.

2. **PP_4790 (lnt / Q88DN4)** — add to satisfy step 3; note and clear the CN-hydrolase superfamily over-propagation flag during review.
3. **PP_0604 (lspA / Q88Q91)** — already the candidate; confirm and retain; broad GO:0004190 worth a note.

(Downstream, for a separate Lol module: PP_4003, PP_0724, PP_2154, PP_2155, PP_2156.)

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8. Key references

- Vogeley L, *et al.* Structural basis of lipoprotein signal peptidase II action and inhibition by globomycin. *Science* 2016.

  P 26912896

  (*P. aeruginosa* LspA structure; aspartyl peptidase.)
- Mao G, *et al.* Crystal structure of *E. coli* lipoprotein diacylglyceryl transferase. *Nat Commun* 2016.

  P 26729647

  (Lgt structure; *lgt* deletion lethal in most Gram-negatives.)
- Noland CL, *et al.* Structural insights into lipoprotein N-acylation by Lnt. *PNAS* 2017.

  P 28698362

  (Sequential three-enzyme maturation; Lnt.)
- Diao J, *et al.* Inhibition of *E. coli* lipoprotein diacylglyceryl transferase… *Antimicrob Agents Chemother* 2021.

  P 33875545

  (Lgt essentiality / OM integrity.)
- Qi HY, *et al.* Structure-function relationship of bacterial prolipoprotein diacylglyceryl transferase. 1995.

  P 7592473

  (Cross-phylum Lgt conservation.)

- Garland K, *et al.* Optimization of globomycin analogs as Gram-negative antibiotics. 2020. P 32768648. (LspA as validated target.)
- Xiao Y, *et al.* Myxovirescin inhibits type II signal peptidase. 2012. P 22232277. (LspA target validation.)

Database evidence: UniProtKB Q88CN8 / Q88Q91 / Q88DN4 (Swiss-Prot, PE3); KEGG ppu:PP_5142/PP_0604/PP_4790 (KO K13292/K03101/K03820); InterPro IPR001640 / IPR001872 / IPR004563; UniPathway UPA00665.